PREPARED BY THEMISIQ COMPLIANCE INC.

Double Materiality Screening Report

CSRD / ESRS — financial and impact materiality screening across the ten ESRS topical standards.

Legal entity	Acme Industries Ltd.
Reporting period	FY2025
Primary sector	Industrials & Manufacturing
Operating regions (IPCC AR6)	Eastern North America (ENA)
Policy jurisdictions	EU (EU ETS), EU CBAM exposure
Scenario tested	IPCC SSP2-4.5 (~2.7°C)
Time horizon	medium term
Asset profile	inland
Model version	0.1-draft
Report date	May 27, 2026

Executive summary

This double materiality screening identifies climate-related risks and sustainability topics that are likely to be material for **Acme Industries Ltd.** under the IPCC SSP2-4.5 pathway over the medium term.

0	1	4
High physical risks	High transition risks	Topics material on both axes

Key findings

- Transition risk: **Technology displacement** is material (high band) under the chosen scenario and jurisdictions.
- E1 **Climate change** is material on **both axes** (financial 7.2 / impact 5.0).
- E2 **Pollution** is material on **both axes** (financial 8.0 / impact 8.0).
- E3 **Water & marine resources** is material on **both axes** (financial 5.0 / impact 8.0).
- E5 **Resource use & circular economy** is material on **both axes** (financial 8.0 / impact 5.0).

Methodology and basis

This screening combines four public, independently-sourced frameworks. No proprietary or licensed third-party classification is reproduced.

Frameworks

- **IPCC AR6 WGI reference regions** — the 20 land regions used here are drawn from the Sixth Assessment Report Working Group I reference-region set (Iturbide et al., 2020), each with its characteristic profile of climatic impact-drivers.
- **IPCC climatic impact-drivers** — the physical hazards (extreme heat, drought, water stress, inland flooding, coastal flooding, wildfire, storms/cyclones, cold/permafrost) follow the AR6 climatic-impact-driver categories.
- **TCFD transition risk categories** — transition risks follow the Task Force on Climate-related Financial Disclosures classification: policy and legal, technology, market, and reputation.
- **ESRS topical standards** — the impact-materiality axis assesses the ten ESRS topical standards (E1–E5 environment, S1–S4 social, G1 governance).
- **Scenario pathways** — both IPCC Shared Socioeconomic Pathways (SSP1-2.6, SSP2-4.5, SSP5-8.5) and NGFS scenarios (Orderly, Disorderly, Hot House) are available; this screening uses IPCC SSP2-4.5.

Risk model

Physical risk is computed as industry sensitivity × regional hazard exposure × scenario severity × time-horizon multiplier. A risk is flagged only where industry sensitivity and regional hazard exposure intersect — preventing the common error of flagging, for example, drought for any agricultural entity regardless of where it operates.

Transition risk is computed as industry carbon exposure × jurisdictional policy intensity × scenario policy-speed × time-horizon multiplier. Transition geography (policy jurisdictions) is deliberately distinct from physical geography (IPCC regions): physical exposure depends on where assets and hazards are; transition exposure depends on which policy regimes apply.

Double materiality is operationalised as the combination of single (financial) materiality and impact materiality: *double materiality = financial + impact*. The financial axis uses the engine above; climate (E1) financial score is taken directly from the physical + transition computation. The impact axis uses industry baselines for the ten ESRS topics, refined by the user's self-assessment.

Scenario inversion

Physical and transition risk move in opposite directions across scenarios: high-warming pathways raise physical risk and lower transition pressure; rapid- or abrupt-policy pathways raise transition risk and lower physical pressure. A resilient strategy must hold up across both ends, which is why both frameworks require testing resilience across a range rather than against a single forecast.

Scenario selection and rationale

Pathway used: IPCC SSP2-4.5 (~2.7°C).

A middle pathway (SSP2-4.5, ~2.7°C) was selected as a reasonable central case for first-pass screening. Higher- or lower-warming pathways can be tested to assess resilience across a range.

Time horizon used: medium term.

A medium-term horizon (to 2040) was selected as the default screening lens. Companies with long-lived physical assets may prefer the long-term view; near-term commitments may benefit from the short-term view.

Under both IFRS S2 and CSRD/ESRS, the choice of scenarios used and the rationale for that choice are themselves disclosable. This section documents that judgment.

Double materiality matrix

Each ESRS topic is plotted on the two axes — financial materiality (vertical) and impact materiality (horizontal). Topics in the top-right quadrant are material on both axes and represent the highest reporting and management priority.

Material on both   Material on one axis   Lower priority

Materiality determination — all topics

All ten ESRS topical standards, with their financial and impact materiality scores (0–10) and band. Sorted by maximum of the two axes.

ESRS topic	Financial	Impact
Pollution	HIGH 8.0	HIGH 8.0
Water & marine resources	MED 5.0	HIGH 8.0
Resource use & circular economy	HIGH 8.0	MED 5.0
Climate change	MED 7.2	MED 5.0
Biodiversity & ecosystems	LOW 2.0	MED 5.0
Own workforce	MED 5.0	LOW 2.0
Workers in the value chain	MED 5.0	LOW 2.0
Business conduct	MED 5.0	LOW 2.0
Affected communities	LOW 2.0	LOW 2.0
Consumers & end-users	LOW 2.0	LOW 2.0

Risk register

Physical risks

Industry sensitivity × regional hazard exposure × scenario × horizon. Flagged only where industry sensitivity meets real regional hazard exposure.

Hazard	Severity	Driving region	Score
Extreme heat	MED	Eastern North America (ENA)	4.8
Inland flooding	MED	Eastern North America (ENA)	4.0

Transition risks

Industry carbon exposure × jurisdictional policy intensity × scenario policy-speed × horizon.

Driver	Severity
Carbon pricing / policy	MED
Market & demand shift	MED
Technology displacement	HIGH
Reputation	MED

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